Resource-Efficient Neural Architectures for Post-Disaster Damage Assessment and Rescue Guidance

CSE499B – Section 13, Group 5

Name	Student ID
Abdullah Al Noman	2022095042
Tamanna Akter Mou	2211951042
Aryan Sami	2231407042
Ridita Afrin Riya	2211622042
Abrar Mohammed Tanzim Alam	2222864042

Proposed Solution

The problem. After a disaster, rescue teams need to know quickly which buildings are damaged or destroyed, how many there are and where to go first, often without internet access. Vision-language models (VLMs), which answer questions about images in plain language, look like a natural fit, but the DisasterM3 benchmark (NeurIPS 2025) shows that even fine-tuned 7-billion-parameter VLMs count damaged buildings poorly, and our own tests confirm it: our fine-tuned Qwen2.5-VL-7B counts at 30.3%, below the base model's 34.2%. BRIGHT (2025) and BDAChat (2026) treat finding a building and grading its damage as separate problems, and TEOChat (ICLR 2025) found that a specialist segmentation network locates buildings far better than a general VLM. Report systems such as DisasTeller rely on an online commercial model, which fails when networks are down. The gap we address is one system that counts from pixels, reasons in language and runs offline on free or small GPUs.

The proposed solution is a hybrid system in which each part does the job it is best at:

1. **Find, then grade.** SAM 3 (Meta, 2025), a general segmentation model prompted with the word "building", outlines every building on the image taken *before* the disaster, where buildings still stand; rubble on the after-image is what our earlier models missed. A grader then labels each outline intact, damaged, destroyed or not a building: a ResNet-18 classifier we trained on SAM 3's outlines of 5,218 training images, which compares a 128 px crop of the building from the before-image with the same crop from the after-image.
2. **Count from a table, not from the VLM.** Every graded building becomes one row of a table (position, size, grade). Counting and location questions are answered by fixed, pre-written queries over this table, following TrustSQL's lesson that a system should refuse rather than answer wrongly: in our tests no small model wrote its own queries well enough (best 73.9% of 297 questions, against a 90% bar), and passing a count through the VLM lost 10 points (27.0% against 37.0% from the table alone).
3. **The VLM for language.** Our fine-tuned 7B, given 28 extra training steps on full-size (1036 px) images, answers scene-level questions such as the disaster type and how objects relate, and writes the report text from the table's facts. It does not grade buildings: on 300 true buildings our grader was right on 212 and the VLM on 100.
4. **An offline report.** The DisasTeller report agents run on our local models, and cite official guidelines (EMS-98, INSARAG, Sphere and others) through a local search that combines meaning and keyword matching, so every recommendation can be traced to its source without internet access. The report comes with the damage map, the counts and a rescue-priority heatmap.
5. **Small enough for the field.** Our U-Net in 8-bit integers is 33 MB and 1.43× faster on a CPU, losing 0.0003 destroyed-building IoU; the 7B is distilled into a 3B model of about 6 GB, following Distilling Step-by-Step; and EfficientSAM3, a lighter SAM 3, is the planned field finder.

Implementation. The damage map came first: ten single-change training runs, then an error analysis showing that 60% of missed destroyed pixels were labelled background, which led to the before-image and to find-then-grade. The table, the counting agent, the 8-bit U-Net, the heatmap, the guideline search and the report agents on our 7B are built and tested; the report's offline proof run, the fixed queries and the 3B remain. Every run has its decision rule written down first, and data splits are checked by image content, which found 40 validation pairs duplicating training images.

Evaluation. Each part is judged against a target fixed in our review report:

- **Damage map:** mean IoU (pixel overlap, from 0 to 1) above 0.52 and destroyed-building IoU above 0.30 on two training seeds, on the 605 clean validation pairs. Find-then-grade currently reaches 0.307 on the mean of three seeds.
- **Counting:** above the base model's 34.2% on DisasterM3's counting questions, with all six of the benchmark's tasks re-scored; we are 1.7 points short.
- **Offline report:** a complete report produced with the internet switched off; the guideline search must return a correct section in its top 3 for at least 24 of 30 fixed test questions (it returns 27).
- **Field model:** size, speed and per-task accuracy of the 3B against the 7B teacher.

Expected outcome. A damage-assessment and rescue-guidance system that finds and grades every building, counts from those grades, writes a cited report, and runs end to end without internet on a free T4 GPU, with a smaller version for field hardware. Its contribution is the design: finding, grading, counting and writing each go to the component that measurably does it best, each choice backed by a test fixed before it ran.